23.2 -Capstone Project Workshop: End-to-End Application Development

Name:V.Sri Charitha

Hno:2303A51626

Smart Student Attendance & Performance Tracker:

Task-1 :Project proposal

Objective:

To develop a web-based application that helps colleges track student attendance, monitor performance, and generate reports using AI-assisted tools.

Features

- Student login & admin login
- Mark attendance (daily/subject-wise)
- View attendance percentage
- Alerts for low attendance
- Performance tracking (marks)
- Dashboard with charts
- Export reports (PDF/Excel)

Tools & Technologies:

- **Frontend:** HTML, CSS, JavaScript (or React)
- **Backend:** Python (Flask) / Node.js
- **Database:** MySQL / SQLite
- **AI Tools:** ChatGPT, GitHub Copilot
- **Deployment:**
  - GitHub Pages (frontend)
  - Render / Heroku (backend)

Expected Output

- A working web application
- Dashboard showing attendance stats
- Admin can manage students
- Deployed live project link

Task 2 – Application Design

Architecture (Simple)

Frontend (HTML/CSS/JS)
↓
Backend API (Flask / Node.js)
↓
Database (MySQL)

Database Tables

- Students (id, name, dept)
- Attendance (id, student_id, date, status)
- Marks (id, student_id, subject, score)

UI Pages

- Login Page
- Dashboard
- Attendance Page
- Reports Page

Task 3 – Development Plan

Phase 1 – Frontend

- Create forms (login, attendance)
- Dashboard UI

Phase 2 – Backend

- API:
  - /login
  - /mark-attendance
  - /get-attendance
  - /get-report

Phase 3 – Integration

- Connect frontend with backend using fetch/AJAX

Task 4 – Documentation

Include in README.md:

- Project description

- Setup steps
- Technologies used
- Screenshots

Task 5 – Testing

Use AI to:

- Generate test cases
- Check errors
- Improve performance

Example:

- Test login validation
- Test attendance calculation

Task 6 – Deployment:

- Upload code to GitHub
- Deploy:
  - Frontend → GitHub Pages
  - Backend → Render / Heroku

```
attendance-tracker/
│
├── backend/
│   ├── app.py
│   ├── database.db (auto created)
│
├── frontend/
│   ├── index.html
│   ├── dashboard.html
│   ├── script.js
│   ├── style.css
│
└── README.md
```

Backend (Flask API): attendance_tracker/backend/

app.py

EXPLORER

AI

frontend

dashboard.html

index.html

script.js

app.py

database.db

app.py | index.html | dashboard.html | script.js

app.py > report

```python
from flask import Flask, request, jsonify
import sqlite3
from flask_cors import CORS

app = Flask(__name__)
CORS(app)

# ---------- DATABASE ----------
def get_db():
    conn = sqlite3.connect("database.db")
    conn.row_factory = sqlite3.Row
    return conn

# ---------- HOME ROUTE (IMPORTANT) ----------
@app.route('/')
def home():
    return "Server Running Successfully!"

# ---------- INIT DATABASE ----------
@app.route('/init')
def init():
    conn = get_db()
    cur = conn.cursor()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

EXPLORER

AI

frontend

dashboard.html

index.html

script.js

app.py

database.db

app.py | index.html | dashboard.html | script.js

app.py > report

```python
def init():

    cur.execute("CREATE TABLE IF NOT EXISTS students(id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT, dept T
    cur.execute("CREATE TABLE IF NOT EXISTS attendance(id INTEGER PRIMARY KEY AUTOINCREMENT, student_id INTE
    cur.execute("CREATE TABLE IF NOT EXISTS marks(id INTEGER PRIMARY KEY AUTOINCREMENT, student_id INTEGER,

    conn.commit()
    conn.close()
    return "Database Ready!"

# ---------- ADD STUDENT ----------
@app.route('/add-student', methods=['POST'])
def add_student():
    data = request.json
    conn = get_db()
    conn.execute("INSERT INTO students(name, dept) VALUES(?,?)",
                 (data['name'], data['dept']))
    conn.commit()
    conn.close()
    return jsonify({"msg": "Student Added"})

# ---------- GET STUDENTS ----------
@app.route('/students')
```

Frontend:
attendance_tracker/frontend/index.html (Add Student Page)

app.py | index.html | dashboard.html | script.js

frontend > index.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Student App</title>
</head>
<body>

<h2>Add Student</h2>

<input id="name" placeholder="Name">
<input id="dept" placeholder="Department">
<button onclick="addStudent()">Add</button>

<br><br>
<a href="dashboard.html">Go Dashboard</a>

<script src="script.js"></script>

</body>
</html>
```

attendance_tracker /frontend/dashboard.html

D: > AI Assisted Coding > attendance_tracker > frontend > dashboard.html > html

```html
<html>
<body>
<h2>Attendance Dashboard</h2>

<button onclick="loadStudents()">Load Students</button>

<ul id="studentList"></ul>

<button onclick="submitAttendance()">Submit Attendance</button>

<h3>Report</h3>
<button onclick="getReport()">Get Report</button>
<ul id="report"></ul>

<script src="script.js"></script>
</body>
</html>
```

attendance_tracker /frontend/script.js

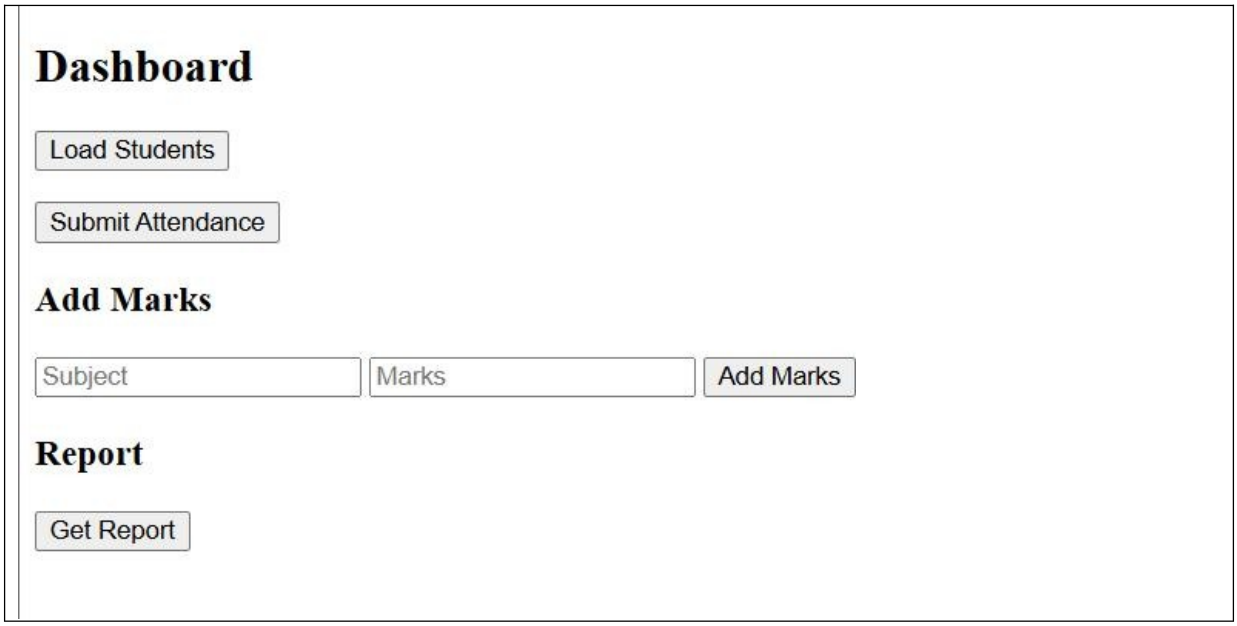
```
app.py    index.html    dashboard.html    script.js
frontend > script.js > ...
const API = "http://127.0.0.1:5000";

function addStudent() {
    const name = document.getElementById("name").value;
    const dept = document.getElementById("dept").value;

    fetch(API + "/add-student", {
        method: "POST",
        headers: {"Content-Type": "application/json"},
        body: JSON.stringify({name, dept})
    })
    .then(res => res.json())
    .then(data => alert(data.msg));
}

function loadStudents() {
    fetch(API + "/students")
    .then(res => res.json())
    .then(data => {
        const list = document.getElementById("list");
        list.innerHTML = "";
```

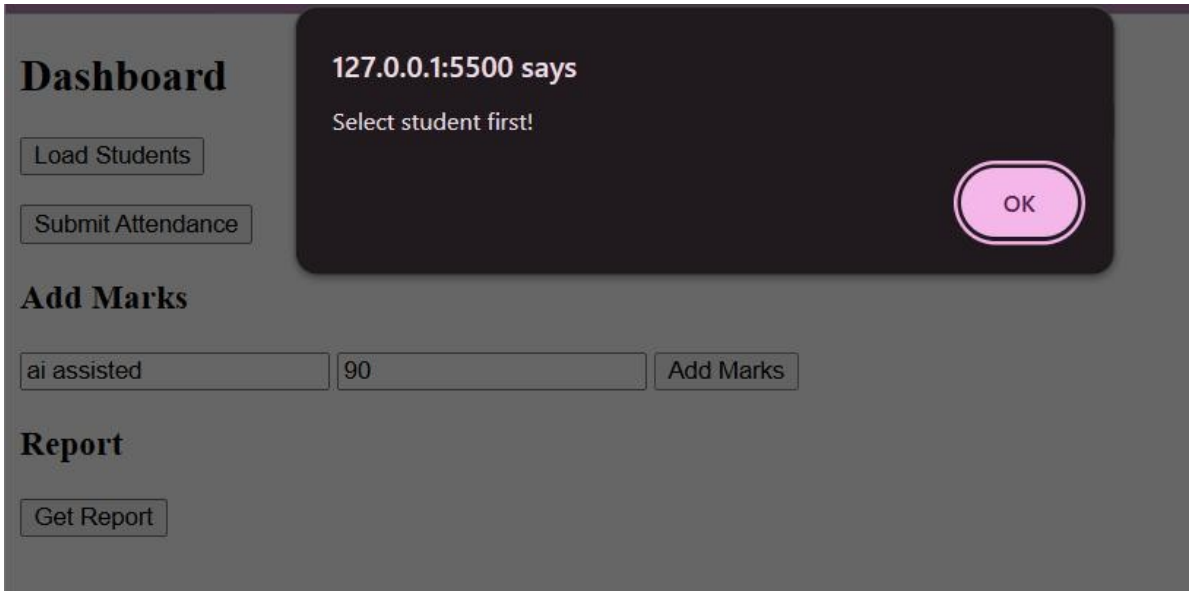
OUTPUT:

Database Ready!

127.0.0.1:5500/frontend/index.html
Add Student
eshwar
cse
Add
Go Dashboard

127.0.0.1:5500/frontend/index.html
Add Student
eshwar
Go Dashboard
127.0.0.1:5500 says
Student Added
OK


Dashboard
Load Students
Submit Attendance
Add Marks
Subject
Marks
Add Marks
Report
Get Report


Dashboard
Load Students
Submit Attendance
Add Marks
ai assisted
90
Add Marks
Report
Get Report
127.0.0.1:5500 says
Select student first!
OK